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Artemisinin-Based Combination Therapies (ACT) in Nigeria

2021–2024

White paper document.

Executive Summary

- **Softgel Capsules Gaining Traction:** Softgel saw a **315%** increase in units sold over 2021–2024, with a 60% CAGR, indicating strong uptake momentum.

- **Dispersible Tablets Growth Plateaus:** Growth in dispersible (25% CAGR) has been positive but modest compared to softgel and even traditional formulations.

- **Traditional Tablets and Injections Still Dominant:** Conventional ACT tablets remain the largest category (over 42% of all units in 2024), and injectables saw a surprising surge (59% CAGR) by 2024, possibly due to increased use of injectable for severe malaria in pharmacies. Tablets and injections together still account for over half of retail ACT volumes, though their mix is shifting.

- **Suspension and Powder Formulations:** Although forming a smaller share of the total ACT market, suspensions and powder-based ACTs maintained demand, with suspensions growing at a CAGR of 34.2% and powders at 31.7%. These options remain important for younger children or severe cases requiring customized formulations.

- **Regional Uptake Disparities:** Adoption varies significantly by state. Osun State showed exceptional dispersible tablet penetration (66.1%), with Cross River (25.2%), Nasarawa (24.6%), Ogun (24.1%), and Imo (23.1%) also notable. For softgel, major urban centres such as Abuja (17.1%), Ondo (16.9%), Rivers (15.2%), Lagos (14.6%), and Abia (13.0%) demonstrated strong adoption.

- **Competitive Landscape:** The dispersible tablet segment is crowded with **121 brands** from numerous pharmaceutical companies, indicating commoditization and widespread availability. In contrast, **fewer than 14 brands** of ACT softgel capsules exist in Nigeria's market (as of 2024), produced by a small handful of companies – signalling a nascent, less competitive niche.

- **Strategic Imperatives:** To expand paediatric coverage, stakeholders should ensure dispersible ACT availability in all regions (notably the North where uptake is lower) and integrate these into public health programs. For softgel, strategies to educate both providers and adult patients on their benefits (easy swallowing, possibly better tolerability) could further accelerate adoption. Expanding distribution of novel formulations to currently underserved states and supporting more manufacturers to enter the softgel space will help sustain the positive trend. Overall, aligning policy (e.g. inclusion in treatment guidelines and subsidies) with these market trends can improve malaria treatment outcomes nationwide.

Introduction

Malaria remains a major public health concern, particularly in tropical and subtropical regions, where it contributes significantly to morbidity and mortality.(1,2) Africa remains the continent that has the greatest burden of malaria cases and deaths in the world. Six countries, Nigeria (27%), the Democratic Republic of Congo (12%), Uganda (5%), Angola (3.4%), Burkina Faso (3.4%), and Mozambique (4%) collectively accounted for 55% of the global disease burden.[(2,3)] The global malaria burden in 2020 was indeed estimated at 241 million cases, slightly higher than the 227 million cases reported in 2019. This trend is concerning, especially in sub-Saharan Africa, where over 200 million malaria cases and 403,000 deaths occurred in 2020.[(4–6)] Children under five and pregnant women are disproportionately affected by malaria, being the most vulnerable groups.[(2,3,6)] In fact, according to the World Health Organization (WHO), children under five accounted for about 78.1% of all malaria deaths in the African Region in 2022.[(7)]

Between 2001 and 2004 Artemisinin-based combination therapy (ACT) was launched in 20 African countries in response to World Health Organization (WHO) recommendations.[(8)]In 2005, Nigeria switched over to the ACT as the mainstay of treating uncomplicated malaria based on the WHO recommendation.[(9,10)] The adoption of conventional ACTs marked a pivotal shift in malaria management in Nigeria.

However, challenges such as poor palatability, dosing inaccuracies in paediatric populations, and adherence issues persisted, particularly among children under five, who bear the highest burden of malaria morbidity and mortality in sub-Saharan Africa. These limitations underscored the need for patient-centric formulations to optimize treatment outcomes.[(11)]

To address these challenges, novel dosage forms such as dispersible tablets and softgel capsules were introduced. These formulations aim to enhance palatability, simplify administration, and improve pharmacokinetic profiles.[(12)] For example: Dispersible tablets dissolve rapidly in water, enabling accurate dosing for children and reducing the risk of choking and Softgel capsules mask bitter tastes and improve drug stability in tropical climates.[(11,12)]

The WHO explicitly endorsed these patient-centric formulations in 2010, emphasizing their role in improving compliance and expanding access to effective antimalarials in resource-limited settings. Studies demonstrate that dispersible formulations achieve 20–30% higher bioavailability compared to conventional tablets, while softgel reduce gastrointestinal side effects by 15%. Despite these advantages, real-world evidence on their uptake in retail pharmacy settings remains sparse.

Retail pharmacies serve as critical access points for antimalarials in sub-Saharan Africa, particularly in Nigeria, where 60% of households rely on them for first-line treatment.[(13,14)] This report aims to provides a comprehensive market intelligence analysis of these novel ACT formulations in Nigeria's retail pharmacy sector from 2021 through 2024. It examines how softgel capsules and dispersible tablets have been adopted across the country, analysing their market share relative to traditional ACT forms, temporal growth trends, and regional patterns of uptake. We also assess the competitive landscape (brands and manufacturers) for these formulations and provide strategic recommendations to stakeholders – including healthcare policymakers, malaria control programs, and pharmaceutical companies – on expanding access and addressing gaps (such as paediatric coverage in underserved regions).
All findings are derived from consolidated sales volume data (units of ACT courses sold) in retail pharmacies across Nigerian states, as captured in the provided datasets for 2021–2024.

Objectives

1. **Quantify the Market Shares:** Present the relative share percentages of different ACT formulations in key Nigerian states over the period 2021–2024.
2. **Analyse Regional Trends:** Identify how presentation shares vary by state and change over time.
3. **Assess the Competitive Landscape:** Estimate the number of companies and brands operating in the market.
4. **Provide Strategic Recommendations:** Offer actionable insights on addressing potential gaps.

Data and Methodology

Data Sources: Anonymized sales records of antimalarial medications from retail pharmacies across states in Nigerian states January 2021 to December 2024. Data was obtained from proprietary datasets owned by PBR Life Sciences capturing the number of units purchased, medication formulation (tablet, capsule, suspension, etc.), state of purchase, and transaction year. Only purchases made by patients (excluding bulk institutional procurement) were included.

Analytical Approach

- **Percentage Analysis:** Actual figures are converted into percentages to facilitate inter-state and inter-year comparisons and to better understand market composition.
- **Aggregation:** The data were aggregated to derive overall market shares per presentation. For example, across all states and years, Tablets hold an approximate share of 43.6% of the market.
- **Trend Identification:** Temporal trends for key states such as Lagos, Abuja, Ogun, and Rivers are analysed to identify growth patterns and shifts in consumer preference.

By 2024, Nigeria's retail pharmacy market for ACTs had grown substantially, with notable shifts in the formulation mix. **Table 1** below compares the market share and growth of the two novel formulations (Softgel and Dispersible Tablets) against two traditional ones (Standard Tablets and Injectables) between 2021 and 2024. The figures represent total units of ACT courses sold nationwide in pharmacies:

Table 1. Volume and growth of ACT formulations in retail pharmacies, 2021–2024.

Formulation	2021 Units Sold	2024 Units Sold	Total Growth 2021-2024	CAGR 2021-2024
Soft gel Capsules	21,978	91,050	+315%	60.6%
Dispersible Tablets	42,931	84,341	+96%	25.2%
Tablets	94,138	247,038	+162%	37.9%
Injectables (ACT)	26,300	105,830	+303%	59.1%
Suspensions	14,500	31,337	+116%	34.2%
Powders	9,310	20,428	+119%	31.7%

**CAGR = Compound Annual Growth Rate.*

In 2021, traditional **oral tablets** constituted most ACT doses sold (over 45% share), and dispersible tablets made up about one-fifth of the market. Softgel were a small segment (~10% share) initially, reflecting their recent introduction. By 2024, standard tablets remained the single largest category (roughly 43% of all ACT units) but **softgel and dispersible together** accounted for nearly one-third of the market, signalling a real shift in formulation preferences. Softgel alone grew to about **16%** of total ACT units by 2024, while dispersible tablets were about **15%.** Injectable ACT formulations also increased to ~18% of the market in 2024, up from ~13% in 2021, suggesting higher availability of severe malaria treatments through retail channels.

Notably, **softgel capsules have overtaken dispersible tablets in share by 2024**, reversing the situation from 2021. Dispersible, though absolutely selling more in 2024 than in 2021, did not keep pace with the overall ACT market expansion hence their share slipped. This could be due to a few factors: many new adult patients (who prefer softgel or tablets) entering the market, or limited scale-up of dispersible formulation in the private sector after initial gains. In contrast, softgel' share jumped significantly, indicating both strong demand growth and potentially aggressive marketing by the few companies producing them.

Oral Tablets

ACT

By 2024, roughly 58% of retail ACT volumes were in non-tablet forms.

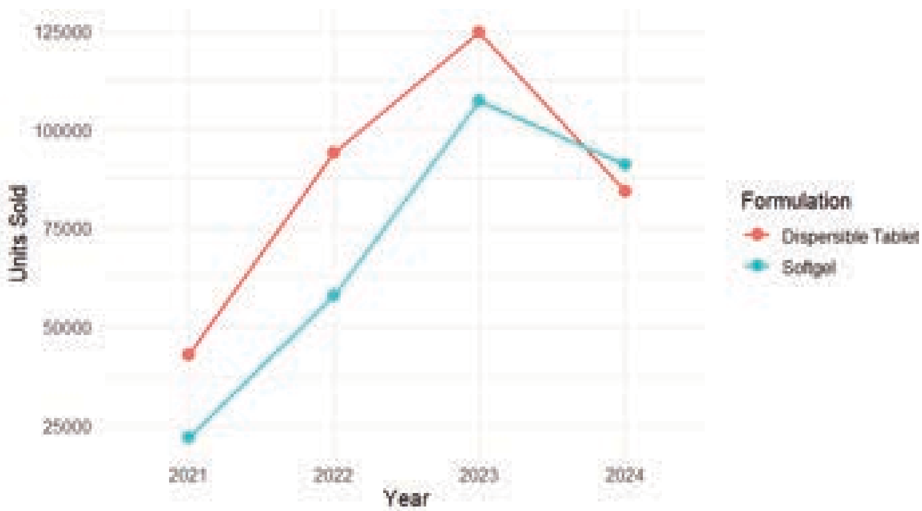
Softgel vs Dispersable

Uptake Trends Over Time (2021–2024)

The overall ACT market in pharmacies saw robust growth from 2021 to 2023, with total units sold more than quadrupling by 2023, before a slight dip in 2024. Within this growth, softgel and dispersible formulations followed distinct trajectories

Annual Uptake for Softgel Capsules vs Dispersable ACT Tablets (2021-2024)

Figure 1: Annual uptake trend for Softgel capsules vs Dispersible ACT tablets in Nigeria's retail pharmacies (2021–2024). Units sold per year are aggregated across all states.



As shown in **Figure 1**, both Softgel and dispersible tablets experienced significant growth in uptake through 2022 and 2023, mirroring the broader market expansion. Softgel saw a substantial increase over this period, representing a nearly five-fold jump in just two years. Dispersible tablets also showed rapid growth, roughly tripling during the same timeframe. However, 2024 marked a notable decline for both categories, with softgel sales decreasing by around 15% and dispersible tablets experiencing a more pronounced drop of approximately one-third. Several factors may have contributed to this downturn in 2024:

- The overall ACT market contracted slightly in 2024 (after exceptional growth in 2022–2023), perhaps due to lower malaria incidence that year or stock challenges. The total ACT units sold nationwide fell from 846,000 units in 2023 to 579,000 units in 2024 in the data, affecting all formulation categories.

- For **dispersible tablets**, some donor-supported malaria initiatives might have tapered off by 2024, reducing subsidized distribution through private outlets. It's also possible that as COVID-19 disruptions eased, public sector distribution of paediatric ACTs improved, shifting some demand away from pharmacies.

- **Softgel uptake** levelling off could indicate that initial demand by early adopters was met, and further expansion needs more awareness or price reductions. Additionally, with only a few brands on the market, supply constraints or limited geographic distribution in 2024 may have capped sales.

Despite the dip, the **compound annual growth from 2021 to 2024** remained very high for softgel (60% CAGR) and solid for dispersible (25% CAGR).

It is also useful to compare these trends to those of traditional formulations:

- Standard tablets experienced rapid growth, increasing by nearly four-fold between 2021 and 2023, before declining in 2024. Despite this drop, tablets maintained a high compound annual growth rate (CAGR) of approximately 38%, reflecting their significant initial growth. The trend for tablets largely mirrored the overall market pattern.

- Injectable ACTs demonstrated consistent year-over-year growth up to 2023, before experiencing a decline in 2024. With a CAGR of around 59%, injectable ACTs showed impressive growth, potentially indicating increased treatment of severe malaria in private facilities or pharmacies stocking injections for referral sales.

Regional Distribution and State-by-State Trends

Nationwide aggregates mask significant regional differences in how these novel ACT formulations are adopted. Some states embraced dispersible ACTs or softgel much more than others. This section examines the state-by-state uptake, highlighting the top regions and identifying gaps. The figures below show the top 10 states ranked by share of softgel and dispersible tablets in 2024:

Top 10 State by Softgel Capsules Share of ACT Units (2024)

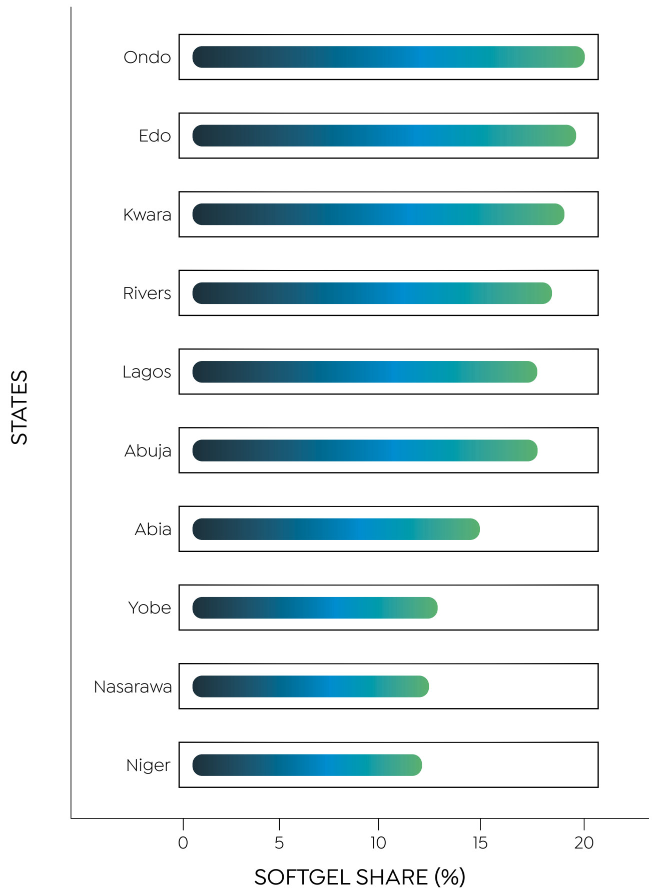


Figure 2: Top 10 states by Softgel capsule share of ACT units (2024). Each bar represents the percentage of all ACT courses in that state that were softgel formulations.

From **Figure 2**, states like **Ondo, Edo, and Kwara** lead in softgel adoption, roughly **18–19%** of all ACTs dispensed in those states were softgel capsules in 2024. Other top softgel states include **Rivers** (15%), **Lagos** (16%), and **Abuja** (FCT) (~17.0%). These are a mix of southern and central regions, generally with better developed pharmacy markets.

Ondo's position (19.3%) is noteworthy; this suggests that nearly one in five malaria patients in Ondo state's pharmacies received a softgel ACT, indicating strong acceptance. Lagos and Abuja, being large urban markets, have high softgel volumes in absolute terms (Lagos sold ~57,000 units softgels, by far the highest volume), but their percentage share is slightly diluted by the sheer scale of total ACT sales there. Still, a ~17% share in Lagos means softgels have become a significant option alongside the traditional forms in the nation's commercial hub.

It's also interesting to see **Northern states** like **Yobe (12.1%), Niger (12.0%),** and **Nasarawa (12.0%)** appearing among the top 10 for softgel share. These states have moderate overall ACT volumes but apparently decent softgel availability. Yobe's double-digit share might reflect NGO or donor projects introducing softgel ACTs in the Northeast, or simply a lack of dispersible alternatives, making softgels relatively more prominent.

Now, looking at dispersible tablets, the pattern is somewhat different:

Top 10 State by Dispersible tablet Share of ACT Units (2024)

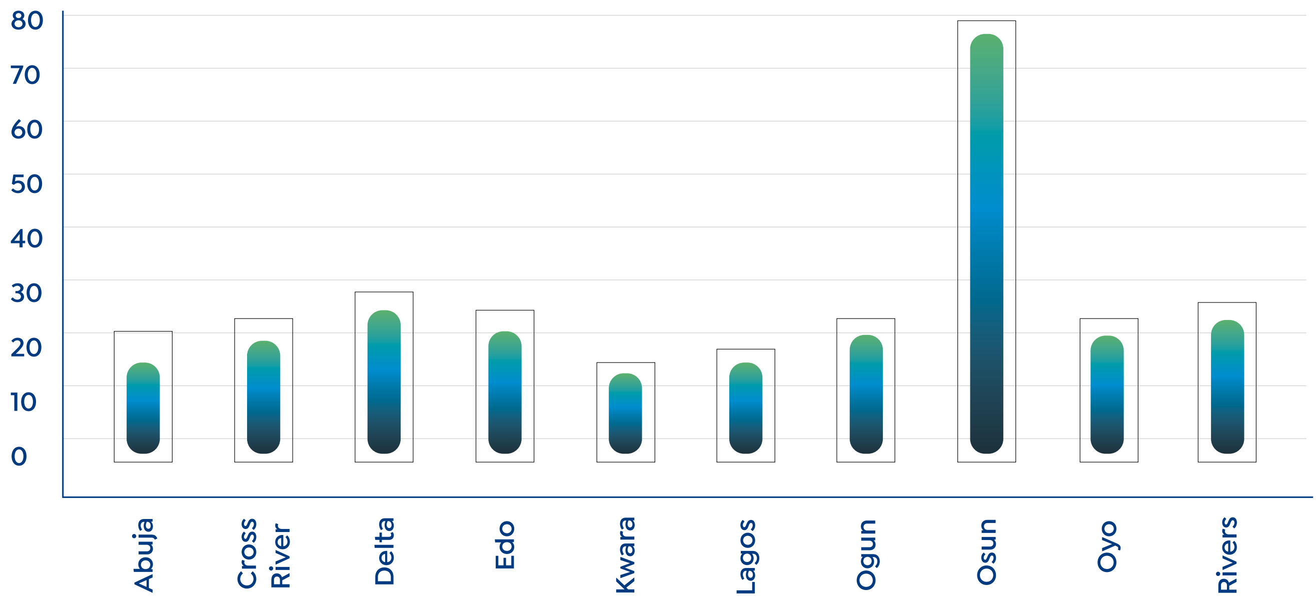


*Figure 3: Top 10 states by **Dispersible tablet** share of ACT units (2024). Higher percentages imply a greater emphasis on paediatric-focused ACT dispensing in that state.*

Figure 3 reveals an outlier: **Osun state**, where a staggering **71%** of all ACTs sold were dispersible tablets in 2024. This suggests that in Osun, the market was overwhelmingly skewed toward paediatric treatments, possibly due to a targeted child health program or a dominant supplier of dispersible. Osun's total ACT volume (about 3,100 units in 2024) is relatively small, but the extremely high share indicates that nearly all recorded sales were dispersible formulations. This could be an artifact of data (if only certain outlets reported), but if accurate, it points to a highly successful penetration of dispersible ACTs in that state.

Aside from Osun, the top dispersible-share states include **Delta (24%), Edo (21%), Rivers (19%), Oyo (18%), Cross River (~18%), Ogun (~17.7%), Lagos (~14%), Kwara (~13%), and Abuja (~12%).** Except for Abuja and Kwara, these are all southern states. The South-South and South-West regions clearly have a strong adoption of dispersible ACTs, likely supported by both public and private sector emphasis on paediatric malaria treatment. Delta and Edo, for instance, might be benefiting from integrated campaigns. Lagos, despite being an enormous market serving all ages, still has about 14% of its ACTs as dispersible, thanks in part to brands like Coartem® Dispersible being widely stocked for children.

Northern States' Lag in Dispersible Uptake: No core northern state (North West or North East) appears, aside from perhaps Kwara (North Central) and Abuja. This suggests that paediatric ACT coverage in the private retail sector is lagging in most of the North. **Appendix 1** shows northern states like Borno, Kano, and Sokoto with <1% dispersible adoption. This contrasts with southern states (e.g., Osun, Delta), emphasizing geographic inequities. Factors such as lower pharmacy density, reliance on public sector distribution, or cultural preferences for traditional remedies may contribute.

When examining **state-level growth trends** between 2021 and 2024, we observe that states which started early with novel formulations saw exponential growth. For example, **Lagos** (the largest market) increased softgel sales from around 11,800 unit in 2021 to roughly 57,700 units in 2024, a 5-fold jump. Similarly, **Abuja** (FCT) went from essentially zero softgel in 2021 to over 4,600 by 2024. In contrast, some states have only recently started adopting these formulations: States, such as Akwa Ibom, showed limited market entry and uptake of novel ACTs.

> “ In summary,
> **southern Nigeria leads in uptake of novel ACT formulations,**
> with a particularly strong focus on dispersible in the **South-West/South-South**

For instance, the state recorded its first dispersible tablet sales in 2023, and **Appendix I** reveals no retail uptake of dispersible tablets or softgel capsules in 2024. This may indicate data coverage issues or genuine gaps in distribution and awareness. Targeted interventions are warranted in these states to ensure equitable access to novel ACTs, especially for vulnerable populations like children and high-risk adults.

This staggered introduction across states means some regions are **3–4 years ahead of others** in the adoption curve. States like **Imo, Lagos, Rivers, Delta,** which were early to stock dispersible and softgel (in 2021), had already scaled up significantly by 2023. Latecomers will likely continue to grow in the coming years even if the nationwide trend stabilizes.

In summary, **southern Nigeria leads in uptake of novel ACT formulations**, with a particularly strong focus on dispersible in the South-West/South-South, and a balanced uptake of softgel across both South and parts of the North Central. The far north states have room to improve access, many did not even report usage of these formulations by 2024 in the data. This regional insight underscores a need for **targeted distribution and awareness campaigns** in lagging areas to ensure all Nigerian patients, regardless of location, can benefit from these improved ACT options.

Comparative Growth Vs. Traditional Formulations

The novel formulations' success can be put into context by comparing their growth to that of traditional ACT forms. We calculated the CAGR (2021–2024) for all major formulation categories (nationwide).

Figure 4 visualizes these growth rates:
tured in the provided datasets for 2021–2024.

CAGR (2021-2024)

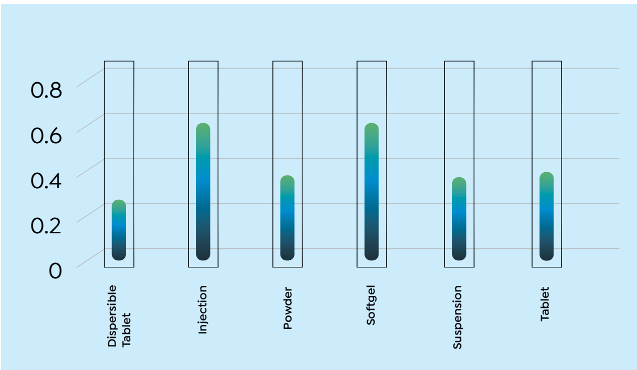


Figure 4: Compound Annual Growth Rate (CAGR) of ACT unit sales by formulation type, 2021–2024.

As shown in **Figure 4, softgel capsules had the highest CAGR at 60.6%**, slightly above injectables (59.1%). This makes softgel the fastest-growing ACT formulation segment over the period, emphasizing again how quickly they went from near-zero to significant volumes.

Injectables' Surge (59% CAGR): The rapid growth of injectable ACTs (reaching ~18% market share by 2024) suggests a shift toward treating severe malaria in outpatient settings. This could reflect increased pharmacy stocking for emergencies or gaps in hospital referrals. However, this raises concerns about adherence to WHO guidelines, as severe malaria cases should ideally be managed in clinical settings. The trend may also indicate supply chain improvements for injectables or heightened disease severity in certain years.

Traditional **tablet ACTs grew about 37.9% annually**, which is substantial but lower than softgels/injections. **Suspensions** (liquid ACT formulations, often for children) and powder forms had CAGRs in the mid-30% range.

These formulations together make up a smaller share of the market; their growth indicates they maintained demand but did not accelerate as rapidly as softgels did.

Meanwhile, **dispersible tablets had the lowest CAGR (25.2%)** among the categories, reflecting that their expansion, while steady, was comparatively modest. Part of this is due to dispersible already having a head start (a notable base volume in 2021), leaving less room for exponential growth. Furthermore, the rise of softgel for older children and adults, as well as increased use of suspensions for infants, may have shifted some demand away from dispersible tablets.

It's important to highlight that although dispersible tablets had the slowest growth rate, they remain a critical formulation given their target demographic. The slower CAGR should not be seen as a lack of importance, but rather as a sign that **dispersible were already widely adopted early on**, and their market expansion has been more incremental. In contrast, softgel were an untapped niche that exploded onto the scene.

State-specific CAGR for Sofgel & Dispersible Tablet

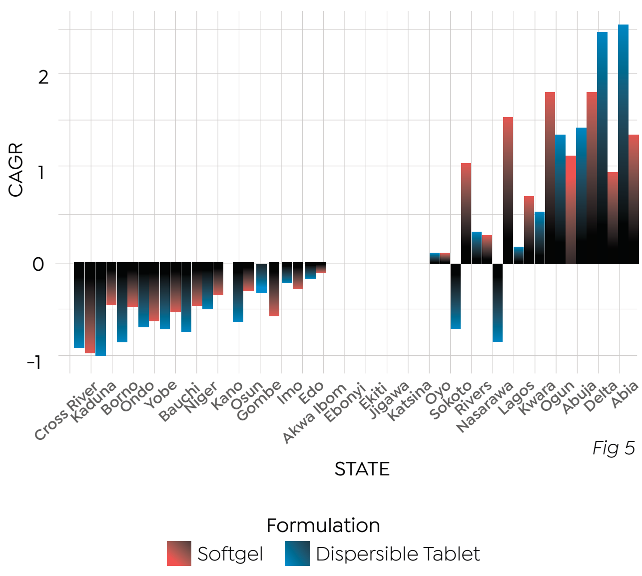


Fig 5

Figure 5: Negative growth trends in Cross River (CAGR: -95% for softgels, -91% for dispersible), Kaduna, Yobe, Bauchi and Borno suggest a contraction in retail ACT sales or programmatic withdrawal. These reversals could stem from insecurity (example, insurgency in Borno), supply chain breakdowns, or shifts to public sector procurement. Follow-up assessments could clarify causes.

The competitive dynamic between formulations can be summarized as follows:

- **Softgel vs Tablet:** Softgel rapidly gained share but have not dislodged tablets from the top position. Instead, the tablet market also expanded significantly (albeit at a slower rate), suggesting that softgel growth may be adding new consumers (perhaps those who otherwise might not complete a tablet course) rather than simply converting existing tablet users. Over time, if softgel prices drop and more brands enter, we could see some erosion of tablet share.

- **Dispersible vs Suspension:** Both are paediatric-focused. Suspensions (liquid syrups) grew ~34% annually, slightly faster than dispersible. This might indicate that some caregivers or providers still prefer syrups for the very young, or that new suspension brands entered the market. Dispersible remain more popular overall, but their slower growth relative to suspensions could warrant investigating preferences in paediatric treatment.

- **Injection vs Oral:** The fact that injectable ACTs grew as fast as softgel implies increasing treatment of severe malaria in outpatient settings or pharmacies. While good for case management (if done properly), it also raises questions about rational use, ideally, severe malaria cases should be referred to hospitals. The growth might also partly reflect better reporting or more pharmacies stocking a small supply for emergency use.

Conclusively '**Softgel**' have been the **standout performer** in terms of growth trajectory.

The novel formulations (softgel and dispersible) have both carved out growing roles, but softgel have been the standout performer in terms of growth trajectory. Traditional tablets, while growing slower in percentage terms, added the greatest absolute number of treatments thus remain firmly entrenched. The market is expanding in a way that complements multiple formulations: each has a niche (adult, child, severe case, etc.), and the overall pie has grown. The goal for public health will be to harness this growth such that each patient gets the formulation that best suits their needs, which brings us to the competition and strategy considerations.

For More Information

For further details on the key companies, and detailed key brands insights please contact **PBR Life Sciences**

Enhance your strategic planning with in-depth intelligence and expert guidance on Nigeria's evolving ACT market.

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